

15. To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs.

To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios.

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int pipefd[2];

    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t p1 = fork();

    if (p1 == 0) {
        int fd = open("result.txt",
                     O_WRONLY | O_CREAT | O_TRUNC,
                     0644);

        if (fd < 0) {
            perror("open");
            exit(1);
        }

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("ls");
        exit(1);
    }
}
```

```
pid_t p2 = fork();

if (p2 == 0) {
    int fd = open("result.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (fd < 0) {
        perror("open");
        exit(1);
    }

    dup2(pipefd[0], STDIN_FILENO);
    dup2(fd, STDOUT_FILENO);

    close(fd);
    close(pipefd[0]);
    close(pipefd[1]);

    execlp("wc", "wc", "-l", NULL);

    perror("wc");
    exit(1);
}

close(pipefd[0]);
close(pipefd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

printf("Result stored in result.txt\n");

return 0;
}
```

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano skill15.c  
bindu@Bindusree:~$ gcc skill15.c -o skill15  
bindu@Bindusree:~$ ./skill15  
Result stored in result.txt  
bindu@Bindusree:~$ |
```